
CO900G - SUSTAINABLE ENGINEERING

IT-BASED SOLUTIONS FOR HIGH ELECTRICITY CONSUMPTION

A Joint Presentation on Smart Monitoring and Peak Power Alerts

Part 1

Manu M Paikada

Part 2

Michelle Jacob

PREVENTIVE APPROACH

THE PEAK POWER ALERT SYSTEM



LIVE MONITORING

Continuous tracking of current household power demand.



EARLY WARNING

Alerts triggered before maximum threshold is reached.



ACTIONABLE ADVICE

Immediate recommendations to reduce unnecessary usage.

HOW IT WORKS

THE 4-STEP ALERT PROCESS

From data input to preventative action.



INPUT

Collect real-time electricity data.



COMPARE

Check demand against threshold.



ALERT

Generate status notification.



RECOMMEND

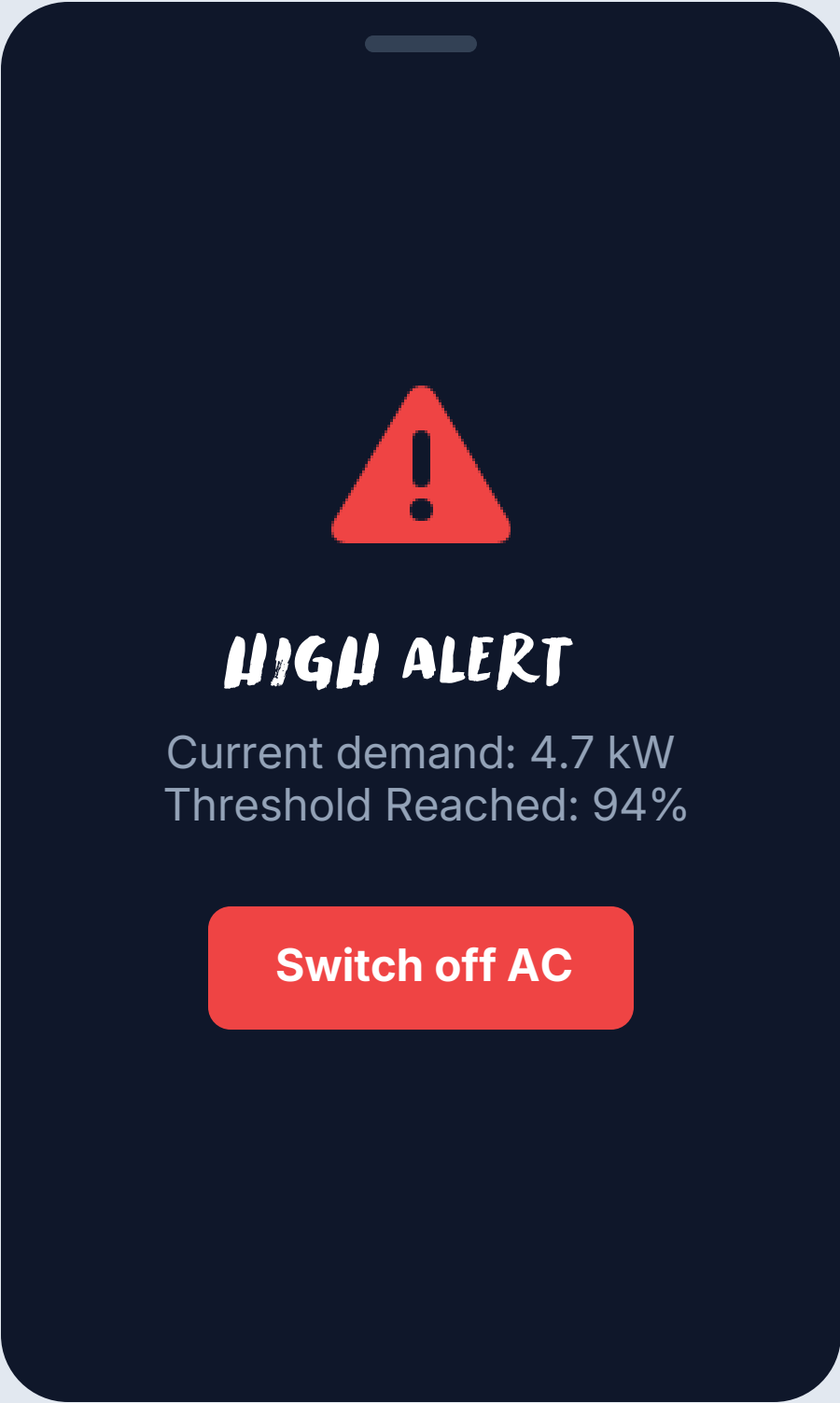
Suggest energy-saving actions.

SYSTEM LOGIC

DYNAMIC STATUS INDICATORS

Clear, actionable alert levels based on custom thresholds.

< 70%	Normal: No warning required.
70 - 90%	Caution: Early warning notification.
90 - 100%	High Alert: Strong warning issued.
> 100%	Peak: Alert plus reduction recommendations.



PROPOSED WEB ARCHITECTURE

1. DASHBOARD

Current power status and system overview.

2. LIVE MONITOR

Real-time gauge against maximum threshold.

3. HISTORY LOGS

Consumption graphs identifying peak periods.

4. ACTIONS MAP

Alerts paired with energy-saving steps.

SUSTAINABILITY GOALS

SDG ALIGNMENT



SDG 7

Affordable and Clean Energy through reduced waste.



SDG 12

Responsible Consumption via informed user decisions.



ISO 14001

Supporting continuous environmental management.



EMPOWERING SUSTAINABLE ENGINEERING

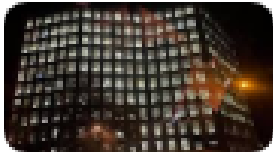
Together, smart monitoring and active alerts create a comprehensive defence against energy waste.

Part 1: Manu M Paikada

Part 2: Michelle Jacob

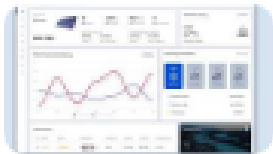
Questions & Discussion

IMAGE SOURCES



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